

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

POSPrinter (TM-T88III/TM-T88IIIX)

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Contents

SECTION 1. INTRODUCTION	1
SECTION 2. DETAILS ON SETTINGS	2
2.1 References of Firmware Versions	2
2.2 Settings of DIP Switches	2
2.3 Port Information	6
2.4 Device Settings	8
2.4.1 Usable Device Specific Settings.....	8
SECTION 3. FUNCTION DETAILS	9
3.1 Property Set Values and Default Values	9
3.1.1 Capability Set Values.....	9
3.1.2 List Properties.....	11
3.1.3 Width and Height Properties	12
3.1.4 Common Property Strings	13
3.1.5 PageMode Print Properties.....	13
3.2 Methods	14
3.3 Escape Sequences	16
3.4 Printable Barcode Type	18
3.5 Power Condition Reports.....	18
3.6 Synchronous Processing	18
3.7 Printing Positions	19
3.8 Electronic Logo Function (NVRAM)	19
3.9 Printable bitmap types and sizes	20
3.10 Maintenance Counter	20
3.11 Automatic Recovery Function	20
3.12 Output without Flow Control on the USB/Ethernet Interfaces	20
SECTION 4. WARNINGS.....	21

Section 1. Introduction

This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-T88III and TM-T88IIIX Series POS Printers are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

Serial	Parallel	USB	Ethernet
TM-T88III	TM-T88IIIP	TM-T88IIIU	TM-T88IIIE
TM-T88IIIM	TM-T88IIIPM	TM-T88IIIMU	TM-T88IIIME
TM-T88IIIX ^{*1} (For IR-700)			
TM-T88IIIXM ^{*1} (For IR-700)			

^{*1} The operation is not supported in Windows Vista environment.

Before reading the manual, see the following explanation about the characteristic of the TM-T88III and TM-T88IIIX models.

- TM-T88III
Station: Receipt (Line Thermal 180 dpi X 180 dpi)
- TM-T88IIIX
Station: Receipt (Line Thermal 180 dpi X 180 dpi)

Throughout the manual, the various model names will be referred to as TM-T88III and TM-T88IIIX.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the TM-T88III and TM-T88III-X Series printers.

2.1 References of Firmware Versions

Refer to the release notes (Relnote.txt).

2.2 Settings of DIP Switches

Confirm that the following settings have been made correctly.

1) Serial port

DIP-SW1

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	ON
8	OFF

Recommended
Recommended
Fixed at OFF
Fixed at OFF
Settable
Settable
Settable
Settable

DIP-SW2

No.	Setting
1	OFF
2	OFF
3	OFF
4	OFF
5	OFF
6	OFF
7	OFF
8	OFF

Settable
Settable
Settable
Settable
Fixed at OFF
Fixed at OFF
Fixed at OFF
Fixed at OFF

It is possible to change the settings of DIP-SW1-1 (Processing of the data input error) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

Set DIP-SW1-3 (Handshake) to DTR/DSR.

Set DIP-SW1-4 (Bit length) to 8 bits.

Set DIP-SW1-5 to DIP-SW1-8 in accordance with the port information.

The described set values are the default values. For the details, refer to the product manual of the POSPrinter. Also, if these settings are changed, make sure to change the port information using the SetupPOS utility.

Set DIP-SW2-2 in accordance with whether or not a customer display is connected. If connected, set to ON. If not, set to OFF.

Set DIP-SW2-3 and DIP-SW2-4 (Specification of the print density) to match the environment of use.

Make other settings in accordance with the settings described above.

- For the TM-T88III and TM-T88IIIX
DIP-SW2-2 is not used. Fix them OFF.

- For the TM-T88III
DIP-SW2-5 can be set.

With DIP-SW1-2 being OFF and DIP-SW2-5 also OFF, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 256 bytes.

With DIP-SW1-2 being OFF but DIP-SW2-5 being ON, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 138 bytes.

If DIP-SW1-2 is ON, then the settings of DIP-SW2-5 become invalid.

2) Parallel Port

DIP-SW 1

No.	Setting	
1	OFF	Recommended
2	OFF	Recommended
3	OFF	Fixed at OFF
4	OFF	Fixed at OFF
5	OFF	Fixed at OFF
6	OFF	Fixed at OFF
7	OFF	Fixed at OFF
8	OFF	Fixed at OFF

DIP-SW 2

No.	Setting	
1	OFF	Settable
2	OFF	Fixed at OFF
3	OFF	Settable
4	OFF	Settable
5	OFF	Fixed at OFF
6	OFF	Fixed at OFF
7	OFF	Fixed at OFF
8	ON	Fixed at ON

It is possible to change the settings of DIP-SW1-1 (Auto line feed) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

Set DIP-SW2-3 and DIP-SW2-4 (Specification of the print density) to match the environment of use.

Make other settings in accordance with the settings described above.

- For the TM-T88III
DIP-SW2-5 can be set.

With DIP-SW1-2 being OFF and DIP-SW2-5 also OFF, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 256 bytes.

With DIP-SW1-2 being OFF but DIP-SW2-5 being ON, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 138 bytes. If DIP-SW1-2 is ON, then the settings of DIP-SW2-5 become invalid.

When parallel I/F is used with Windows 2000, Windows XP or Windows Vista, please set DIP-SW2-1 (BUSY condition) to ON (Receive buffer full).

3) USB Port

DIP-SW1			DIP-SW2		
No.	Setting		No.	Setting	
1	OFF	Recommended	1	OFF	Recommended
2	OFF	Recommended	2	OFF	Fixed at OFF
3	OFF	Fixed at OFF	3	OFF	Settable
4	OFF	Fixed at OFF	4	OFF	Settable
5	OFF	Fixed at OFF	5	OFF	Fixed at OFF
6	OFF	Fixed at OFF	6	OFF	Fixed at OFF
7	OFF	Fixed at OFF	7	OFF	Fixed at OFF
8	OFF	Fixed at OFF	8	ON	Fixed at ON

It is possible to change the settings of DIP-SW1-1 (Auto line feed) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

Set DIP-SW2-3 and DIP-SW2-4 (Specification of the print density) to match the environment of use.

Make other settings in accordance with the settings described above.

- For the TM-T88III

DIP-SW2-5 can be set.

With DIP-SW1-2 being OFF and DIP-SW2-5 also OFF, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 256 bytes.

With DIP-SW1-2 being OFF but DIP-SW2-5 being ON, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 138 bytes.

If DIP-SW1-2 is ON, then the settings of DIP-SW2-5 become invalid.

4) Ethernet Port

DIP-SW1

No.	Setting	
1	OFF	Recommended
2	OFF	Recommended
3	OFF	Fixed at OFF
4	OFF	Fixed at OFF
5	OFF	Fixed at OFF
6	OFF	Fixed at OFF
7	OFF	Fixed at OFF
8	OFF	Fixed at OFF

DIP-SW2

No.	Setting	
1	OFF	Recommended
2	OFF	Fixed at OFF
3	OFF	Settable
4	OFF	Settable
5	OFF	Fixed at OFF
6	OFF	Fixed at OFF
7	OFF	Fixed at OFF
8	ON	Fixed at ON

It is possible to change the settings of DIP-SW1-1 (Auto line feed) and DIP-SW1-2 (Specification of the received buffer capacity), but it is recommended to leave them OFF.

Set DIP-SW2-3 and DIP-SW2-4 (Specification of the print density) to match the environment of use.

Make other settings in accordance with the settings described above.

- For the TM-T88III

DIP-SW2-5 can be set.

With DIP-SW1-2 being OFF and DIP-SW2-5 also OFF, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 256 bytes.

With DIP-SW1-2 being OFF but DIP-SW2-5 being ON, the receive buffer is full if the free space of the receive buffer is from 128 bytes to 138 bytes.

If DIP-SW1-2 is ON, then the settings of DIP-SW2-5 become invalid.

2.3 Port Information

1) Port information when serial port is used

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Baud rate [bps]	4800, 9600, 19200, 38400
Bit length [bit]	8
Parity	NONE, ODD, EVEN
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Baud rate [bps]	9600 (TM-T88III) 38400 (TM-T88IIIX)
Bit length [bit]	8
Parity	NONE
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	1024
Output interval time [ms]	2500

2) Port information when using parallel port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	1024
Output interval time [ms]	2500

3) Port information when using USB port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output interval time [ms]	0 to 9999

The default setting is as shown in the following table.

Item	Setting range
Output interval time [ms]	2500

4) Port information when using Ethernet port

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Output buffer length [byte]	1024
Output interval time [ms]	2500

2.4 Device Settings

The following explanation is about the settings for TM-T88III and TM-T88IIIX.

2.4.1 Usable Device Specific Settings

For the TM-T88III and TM-T88IIIX, the following device specific settings are settable by the SetupPOS utility. For the detail, please refer to the corresponding part of the Section 2 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”

Tab	Settings
General	Disable panel buttons
	Assume print complete when data output finishes
	Homogenize Error Codes ^{*1}
	Ignore firmware version check
	Output complete timeout
Bitmap	TMFlogo...
Color Bitmap	Halftone: Method
	Halftone: Brightness
	Color: Primary
Status Log	ERROR
	OFFLINE
	Log file name (full path name)
	Maximum file size [KB]
Default Value ^{*2}	Multilingual font
Printing Properties ^{*3}	Receipt Characters per Line
	Receipt Line Spacing [dots]
	CharacterSet [CodePage Number]

^{*1} The settings can be changed when using a connection other than serial.

^{*2} Available only for the Multilingual character model.

^{*3} Available only for the TM-T88III and TM-T88IIIM.

Section 3. Function Details

This section describes the functions of the TM-T88III and TM-T88IIIX printers in details. Supplementary explanation of the parts not described in detail in the "UPOS" is also given here.

3.1 Property Set Values and Default Values

The following explanation is about the property set values and the default values.

3.1.1 Capability Set Values

The following values are the Capability set values.

Capability Name	Setting Value	
	TM-T88III/TM-T88IIIM	TM-T88IIIX/TM-T88IIIXM
CapTransaction	TRUE	TRUE
CapCoverSensor	TRUE	TRUE
CapConcurrentRecSlp	FALSE	FALSE
CapConcurrentJrnSlp	FALSE	FALSE
CapConcurrentJrnRec	FALSE	FALSE
CapConcurrentPageMode	FALSE	FALSE
CapCharacterSet ^{*1}	PTR_CCS_UNICODE	PTR_CCS_ASCII
CapMapCharacterSet	TRUE ^{*2}	FALSE
CapJrnUnderline	FALSE	FALSE
CapJrnNearEndSensor	FALSE	FALSE
CapJrnItalic	FALSE	FALSE
CapJrnEmptySensor	FALSE	FALSE
CapJrnDwideDhigh	FALSE	FALSE
CapJrnDwide	FALSE	FALSE
CapJrnDhigh	FALSE	FALSE
CapJrnColor	0	0
CapJrnCartridgeSensor	0	0
CapJrnBold	FALSE	FALSE
CapJrn2Color	FALSE	FALSE
CapJrnPresent	FALSE	FALSE
CapRecPageMode	TRUE	TRUE
CapRecUnderline	TRUE	TRUE
CapRecStamp	FALSE	FALSE
CapRecRotate180	TRUE	TRUE
CapRecRight90	TRUE	TRUE

CapRecPapercut	TRUE	TRUE
CapRecNearEndSensor	TRUE	TRUE
CapRecMarkFeed	0	0
CapRecLeft90	TRUE	TRUE
CapRecItalic	FALSE	FALSE
CapRecEmptySensor	TRUE	TRUE
CapRecDwideDhigh	TRUE	TRUE
CapRecDwide	TRUE	TRUE
CapRecDhigh	TRUE	TRUE
CapRecColor	PTR_COLOR_PRIMARY	PTR_COLOR_PRIMARY
CapRecCartridgeSensor	0	0
CapRecBold	TRUE	TRUE
CapRecBitmap	TRUE	TRUE
CapRecBarCode	TRUE	TRUE
CapRec2Color	FALSE	FALSE
CapRecPresent	TRUE	TRUE
CapRecRuledLine	FALSE	—
CapSlpUnderline	FALSE	FALSE
CapSlpRotate180	FALSE	FALSE
CapSlpRight90	FALSE	FALSE
CapSlpNearEndSensor	FALSE	FALSE
CapSlpLeft90	FALSE	FALSE
CapSlpItalic	FALSE	FALSE
CapSlpEmptySensor	FALSE	FALSE
CapSlpDwideDhigh	FALSE	FALSE
CapSlpDwide	FALSE	FALSE
CapSlpDhigh	FALSE	FALSE
CapSlpColor	0	0
CapSlpCartridgeSensor	0	0
CapSlpBothSidesPrint	FALSE	FALSE
CapSlpBold	FALSE	FALSE
CapSlpBitmap	FALSE	FALSE
CapSlpBarCode	FALSE	FALSE
CapSlp2Color	FALSE	FALSE
CapSlpFullslip	FALSE	FALSE
CapSlpPresent	FALSE	FALSE
CapSlpPageMode	FALSE	FALSE
CapSlpRuledLine	FALSE	—

*1 If Multilingual character model, "PTR_CCS_KANJI" is set.

*2 If Multilingual character model, "FALSE" is set.

3.1.2 List Properties

The List Properties are explained in the following.

List Property	Settings	
	TM-T88III/TM-T88IIIM	TM-T88IIIX/TM-T88IIIXM
CharacterSetList	"255,437,850,852,858,860, 863,865,866,998,999,1252" *1*2	"255,437,850,852,858,860, 863,865,866,998,999,1252" *1
JrnLineCharsList	{} ¹	{} ¹
RecLineCharsList (When 79.5 mm is set)	"42,56"	"42,56"
RecLineCharsList (When 57.5 mm is set)	"30,40"	"30,40"
SlpLineCharsList	{} ¹	{} ¹
RecBarCodeRotationList	"0,R90, L90, 180"	"0,R90, L90, 180"
RecBitmapRotationList	"0,R90, L90, 180"	"0,R90, L90, 180"
SlpBarCodeRotationList	{} ¹	{} ¹
SlpBitmapRotationList	{} ¹	{} ¹
FontTypefaceList	{} ¹	{} ¹

*1 If Multilingual character model, "936" or "950" is added to the list.

*2 When the CapCharacterSet property is set to "PTR_CCS_UNICODE," "997" is added to the list. When CharacterSet is set to "997," all characters loaded in the device are allocated to Unicode for printing. However, the BinaryConversion property should be set to "OPOS_BC_NONE" when printing with Unicode.

3.1.3 Width and Height Properties

The width and height properties are described below.

Property	Settings		
	Default Value	Maximum value [dot]	Minimum value [dot]
RecLineSpacing	30	127	24 ^{*1}
JrnLineSpacing	X	X	X
SlpLineSpacing	X	X	X
SlpLineHeight [dot]	9		
RecLineHeight [dot]	24,17		
JrnLineHeight [dot]	X		
SlpLineWidth [dot]	X		
RecLineWidth [dot]	512		
JrnLineWidth [dot]	X		
RecSidewaysMaxLines	17 ^{*2}		
RecSidewaysMaxChars (When Font A is selected)	69		
RecSidewaysMaxChars (When Font B is selected)	92		
RecLinesToPaperCut	5 ^{*3}		
SlpSidewaysMaxLines	X		
SlpSidewaysMaxChars	X		
SlpMaxLines	X		

X: No settings

^{*1} In the case of a line thermal station, the Line Spacing setting is identical with the height of the characters which means that it can be set at up to 17 when Font B is selected.

^{*2} It can be changed by the settings of the RecLineSpacing or the RecLineHeight.

^{*3} It can be changed by the settings of the RecLineSpacing or the character height.

3.1.4 Common Property Strings

The Device information properties are described below.

- TM-T88III

I/F	DeviceName	DeviceDescription
S	TM-T88III	EPSON TM-T88III POS Printer
	TM-T88IIIM	EPSON TM-T88IIIM POS Printer
P	TM-T88IIIP	EPSON TM-T88IIIP POS Printer
	TM-T88IIIPM	EPSON TM-T88IIIPM POS Printer
U	TM-T88IIIU	EPSON TM-T88IIIU POS Printer
	TM-T88IIIMU	EPSON TM-T88IIIMU POS Printer
E	TM-T88IIIE	EPSON TM-T88IIIE POS Printer
	TM-T88IIIME	EPSON TM-T88IIIME POS Printer

- TM-T88IIIX

I/F	DeviceName	DeviceDescription
S	TM-T88IIIX	EPSON TM-T88IIIX POS Printer
	TM-T88IIIXM	EPSON TM-T88IIIXM POS Printer

I/F indicate the connected interface.

The following is the list of the four connecting interfaces.

S: Serial

P: Parallel

U: USB

E: Ethernet

3.1.5 PageMode Print Properties

The Device information properties are described below.

- TM-T88III

Property	Station ^{*2}		
	Journal	Receipt	Slip
PageModeArea	-	(80mm)"512", "831" (58mm)"360", "831"	-
PageModeDescriptor ^{*1}	-	BM/BC/BMR/BCR	-

- TM-T88IIIX

Property	Station ^{*2}		
	Journal	Receipt	Slip
PageModeArea	-	"512", "831"	-
PageModeDescriptor ^{*1}	-	BM/BC/BMR/BCR	-

^{*1} Following setting values are used for the PageModeDescriptor property.

BM : Bitmap printing is available.

BC : Barcode printing is available.

BMR : Rotated printing of bitmap is available.

BCR : Rotated printing of barcode is available.

^{*2} If the Station's CapRecPageMode property value is FALSE, the PageModeArea property shall have "" and the PageModeDescriptor property shall have "0" respectively as a setting value.

3.2 Methods

The following explanation is about supported/unsupported Methods, and the detailed information.

TM-T88III/TM-T88IIIM:

Method	Supported/Unsupported	Compatibility with the PageMode printing
PrintNormal	O	O
PrintTwoNormal	X	X
PrintImmediate	O	O ^{*1}
PrintBarCode	O	O ^{*2}
PrintBitmap	O	O ^{*3}
PrintMemoryBitmap	O	O ^{*3}
CutPaper	O (1~100: Cutting with one point of the bottom left corner uncut)	X
MarkFeed	X	X
ChangePrintSide	X	X
ValidateData	O	O
TransactionPrint	O	O
SetLogo	O	O
SetBitmap	O	O
RotatePrint	O	X
EndRemoval	X	X
BeginRemoval	X	X
EndInsertion	X	X
BeginInsertion	X	X
ClearPrintArea	O	O
PageModePrint	O	O
DrawRuledLine	X	X

O:Supported

X:Unsupported

^{*1} If the specified Station is ready to print, the printing data shall not be stored in the PageMode printing buffer but, instead, go straight to printing. If the Station is not ready to print, an error is returned.

^{*2} If other than "LEFT" is specified for the printing position of barcode, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

^{*3} If other than "LEFT" is specified for the printing position of bitmap, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

TM-T88IIIX/TM-T88IIIXM:

Method	Supported/Unsupported	Compatibility with the PageMode printing
PrintNormal	O	O
PrintTwoNormal	X	X
PrintImmediate	O	O ^{*1}
PrintBarCode	O	O ^{*2}
PrintBitmap	O	O ^{*3}
PrintMemoryBitmap	O	O ^{*3}
CutPaper	O (1~100: Cutting with one point of the bottom left corner uncut)	X
MarkFeed	X	X
ChangePrintSide	X	X
ValidateData	O	O
TransactionPrint	O	O
SetLogo	O	O
SetBitmap	O	O
RotatePrint	O	X
EndRemoval	X	X
BeginRemoval	X	X
EndInsertion	X	X
BeginInsertion	X	X
ClearPrintArea	O	O
PageModePrint	O	O

O:Supported

X:Unsupported

^{*1} If the specified Station is ready to print, the printing data shall not be stored in the PageMode printing buffer but, instead, go straight to printing. If the Station is not ready to print, an error is returned.

^{*2} If other than "LEFT" is specified for the printing position of barcode, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

^{*3} If other than "LEFT" is specified for the printing position of bitmap, the printing shall be done, regardless of the PageModeHorizontalPosition property setting, based on the PageModePrintArea property setting in the horizontal direction.

3.3 Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

TM-T88III/TM-T88IIIM:

Escape Sequence	Supported/Unsupported	Compatibility with the PageMode printing
#P	0~100	X
#fP	0~100	X
#sP	X	X
sL	X	X
#B	O	O
tL	O	O
bL	O	O
[*]#R	O	O
#lF	0~9999	O
#uF Base Pitch [inch]	0~ equiv. 50 cm	O
#rF Maximum [inch]	X	X
[*]#E	0~65535	X
#fT	X	X
bC	O	O
!bC	O	O
#uC	1~2	O
iC	X	X
!iC	X	X
#rC	1	O
rvC	O	O
!rvC	O	O
#sC	X	X
#fC	X	X
tbC	X	X
!tbC	X	X
tpC	X	X
!tpC	X	X
1C	O	O
2C	O	O
3C	O	O
4C	O	O
#hC	1~8	O
#vC	1~8	O
cA	O	O ^{*1}
rA	O	O ^{*1}
lA	O	O
#stC	1	1
!stC	O	O
*#dL	X	X
N	O	O

O :Supported

X :Unsupported

Numbers: Settable range

^{*1} Regardless of the PageModeHorizontalPosition property setting, center or right adjust

what is to be printed based on the PageModePrintArea property setting in the horizontal direction.

TM-T88IIIX/TM-T88IIIXM:

Escape Sequence	Supported/Unsupported	Compatibility with the PageMode printing
#P	0~100	X
#fP	0~100	X
#sP	X	X
sL	X	X
#B	O	O
tL	O	O
bL	O	O
#R	O	O
#lF	0~9999	O
#uF Base Pitch [inch]	0~ equiv. 50 cm	O
#rF Maximum [inch]	X	X
#E	0~65535	X
#fT	X	X
bC	O	O
!bC	O	O
#uC	1~2	O
iC	X	X
!iC	X	X
#rC	1	O
rvC	O	O
!rvC	O	O
#sC	X	X
#fC	X	X
tbC	X	X
!tbC	X	X
tpC	X	X
!tpC	X	X
1C	O	O
2C	O	O
3C	O	O
4C	O	O
#hC	1~8	O
#vC	1~8	O
cA	O	O ^{*1}
rA	O	O ^{*1}
lA	O	O
N	O	O

O :Supported

X :Unsupported

Numbers: Settable range

^{*1} Regardless of the PageModeHorizontalPosition property setting, center or right adjust what is to be printed based on the PageModePrintArea property setting in the horizontal direction.

3.4 Printable Barcode Type

The TM-T88III and TM-T88IIIX allow the following barcode types.

- Code 128
- Code 128 Parsed
- Code 93
- Codabar
- ITF
- Code 39
- JAN 13 (EAN 13)
- JAN 8 (EAN 8)
- UPC-E
- UPC-A

3.5 Power Condition Reports

The TM-T88III and TM-T88IIIX support Power Condition Reports as follows.

Powered on reporting: Supported

Powered off reporting: Unsupported

3.6 Synchronous Processing

The TM-T88III and TM-T88IIIX use Process ID to determine output completion.

Use of the Process ID allows multiple print commands to be queued to the printer simultaneously. For this reason, Asynchronous output (AsyncMode = TRUE) gives a performance improvement.

3.7 Printing Positions

The TM-T88III and TM-T88IIIX support the function for setting printing position.

Function	Receipt
Left margin	O
Printing Position	O

O: Supported

X: Unsupported

When the left margin setting function is supported, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When the printing position settings are supported, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

3.8 Electronic Logo Function (NVRAM)

The TM-T88III and TM-T88IIIX models feature an electronic logo function (NVRAM). To use NVRAM, start up TMFlogo utility from “Device Specific Settings” of SetupPOS utility, and register image files (BMP style) with NVRAM in advance.

For the details of the registration, please refer to the “Help” of “TMFlogo utility” and/or “EPSON OPOS ADK MANUAL User’s Manual TMFlogo Utility”.

To print image files registered with NVRAM, please use the either of the following

DirectIO:

PTR_DI_FLASH_BITMAP

PTR_DI_FLASH_BITMAP2.

Please refer to the corresponding part of the Section 4 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)” for detail. The available NVRAM sizes are as follows:

TM-T88III : 262144 bytes

TM-T88IIIX : 262144 bytes

3.9 Printable bitmap types and sizes

The TM-T88III and TM-T88IIIX support the following bitmap commands. For the detail, please refer to the corresponding part of the Section 3 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”.

The allowance ranges for bitmaps are as follows.

Bitmap command type	Allowance range		
Download bitmap	X (dot)	y (dot)	xy
	1~2040	1~384	<= 98304
Raster bitmap	1~524280	1~2303	
One-line bitmap	No setting range		

3.10 Maintenance Counter

The TM-T88III and TM-T88IIIX do not support the Maintenance Counter.

3.11 Automatic Recovery Function

The TM-T88III and TM-T88IIIX models feature a function for automatic recovery when the power is turned on again after an interruption of power. Recovery processing is performed automatically when the printer's power is turned on again after an interruption. The recovery processing restores the printer to the condition it was in before the power was turned off.

3.12 Output without Flow Control on the USB/Ethernet Interfaces

The TM-T88III and TM-T88IIIX support outputting without flow control on the USB/Ethernet interfaces. The operations differ by the firmware versions. See the corresponding part of the section 2 of this manual.

Section 4. Warnings

This section describes precautions in use of TM-T88III and TM-T88IIIX.

- Because of the method of devices, TM-T88III and TM-T88IIIX are unable to operate 90-degree rotated printing of Raster bitmap. To operate 90-degree rotated printing of a Bitmap, please use `PrintBitmap` method. Another way to operate is to execute `PTR_DI_SET_BITMAP_MODE PTR_DI_BMP_NORMAL DirectIO` and change the setting to use 1-line Bitmap before executing `SetBitmap` method.
Download bitmaps are able to operate 90-degree rotated printing regardless of the setting of `PTR_DI_SET_BITMAP_MODE DirectIO`.
- To replace heretofore used TM-T88III (80 mm) with TM-T88III (58 mm), delete the registration of the device first, and then re-register on SetupPOS utility.